

Notes: Conlon and Gortmaker (J. Econometrics, 2025)

Incorporating Micro Data into Differentiated Products Demand Estimation with PyBLP

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1 Big picture

- **Question.** How should researchers incorporate individual-level consumer data into Berry, Levinsohn, and Pakes-style differentiated-products demand estimation? The paper answers this as both an econometric question and a practical PyBLP implementation question.
- **Main object.** The paper studies a *micro BLP estimator*: the usual aggregate BLP GMM moments based on product-market shares and instruments are augmented with extra moments based on surveys, consumer panels, second choices, or summary statistics.
- **Motivation.** Aggregate BLP can estimate rich substitution patterns only when aggregate data contain enough variation in product assortments, prices, demographics, and instruments. In many empirical settings, this variation is weak, so flexible heterogeneous-taste parameters are noisy or poorly identified.
- **Central idea.** Micro data provide within-market variation linking consumer characteristics to choices. This is exactly the variation that helps estimate how demographics and unobserved tastes shift substitution patterns.
- **Framework contribution.** The paper gives a standardized notation for micro datasets, sampling probabilities, micro values, micro parts, and micro moments. This is meant to unify many empirical papers that previously implemented micro BLP in paper-specific ways.
- **Practical contribution.** The framework is implemented in PyBLP, extending Conlon and Gortmaker's earlier recommended practices for aggregate BLP estimation to the case with micro data.
- **Micro moments.** The paper supports a broad class of moments: probabilities, conditional means, covariances, correlations, regression coefficients, weighted survey statistics, and second-choice moments.
- **Optimal micro moments.** If the researcher has complete micro data, the paper shows how to use score-based moments. These moments are optimal in the Chamberlain sense within the class of micro BLP estimators considered.
- **Key warning.** More micro data are not automatically better. If micro data are sampled from a population that is incompatible with the aggregate data, or if variables are measured differently, adding micro moments can worsen estimates.
- **Monte Carlo message.** Well-targeted standard micro moments substantially improve finite-sample performance, and optimal micro moments improve further when the micro data are complete and compatible. Second-choice data are especially valuable for estimating unobserved taste heterogeneity.
- **Empirical example.** The paper estimates demand for soft drinks in Seattle using NielsenIQ scanner data, NielsenIQ panel data, and a small online second-choice survey. It then predicts substitution under Seattle's sweetened beverage tax.
- **Takeaway.** The paper is best read as a practical econometric guide: use aggregate BLP carefully, diagnose whether aggregate variation is enough, add micro data when it targets the weakly identified taste parameters, test compatibility, and use PyBLP to compute moments, weighting matrices, standard errors, and optimal moments.

2 Data, notation, and measurement

2.1 Aggregate data

The aggregate part is the standard BLP environment. Markets are indexed by

$$t \in \mathcal{T}, \quad (1)$$

and each market has a finite choice set of inside products

$$j \in \mathcal{J}_t, \quad (2)$$

plus an outside option denoted by $j = 0$.

For each product-market observation, the researcher observes

$$(x_{jt}, z_{jt}, \mathcal{S}_{jt}), \quad (3)$$

where x_{jt} are demand characteristics, z_{jt} are instruments, and $\mathcal{S}_{jt}$ is the observed market share.

The key unobservable is product quality:

$$\xi_{jt}. \quad (4)$$

The aggregate identifying moments are based on the usual exclusion restriction

$$\mathbb{E}[\xi_{jt} z_{jt}] = 0. \quad (5)$$

Interpretation. Aggregate data identify mean utilities and the relationship between product characteristics and unobserved quality. But aggregate data alone may not contain enough variation to sharply identify heterogeneous tastes.

2.2 Consumer types and demographics

The model represents consumers in each market by discrete types

$$i \in \mathcal{I}_t, \quad (6)$$

with weights

$$w_{it} \in [0, 1], \quad \sum_{i \in \mathcal{I}_t} w_{it} = 1. \quad (7)$$

Each type has observed demographics and unobserved taste shocks:

$$y_{it} \in \mathbb{R}^R, \quad \nu_{it} \in \mathbb{R}^C. \quad (8)$$

The consumer-type representation is both notational and computational. In practice, i often indexes Monte Carlo draws, quadrature nodes, or survey-weighted demographic types.

Interpretation. The weights w_{it} describe the population distribution of consumers in the aggregate demand model. They are not the same thing as survey sampling weights.

2.3 Micro datasets

Micro datasets are indexed by

$$d \in \mathcal{D}. \quad (9)$$

A micro dataset is generated by a survey or panel that samples some consumers from the aggregate population. A sampled consumer is described by a tuple

$$(t_n, i_n, j_n), \quad (10)$$

where t_n is the market, i_n is the consumer type, and j_n is the chosen product.

The sampling probability is written

$$\mathbb{P}_A(n \in \mathcal{N}_d \mid t_n = t, i_n = i, j_n = j) = w_{dijt}. \quad (11)$$

This allows the survey to be selected by market, demographic type, purchase status, product group, brand, or any combination of these.

Examples:

$$w_{dijt} \propto \mathcal{M}_t \quad \text{random sample proportional to market size,} \quad (12)$$

$$w_{dijt} \propto \mathbf{1}\{j \neq 0\} \quad \text{sample of inside-good purchasers,} \quad (13)$$

$$w_{dijt} \propto \mathbf{1}\{j \in \mathcal{J}_b\} \quad \text{sample of buyers of brand or group } b, \quad (14)$$

$$w_{dijt} \propto \mathbf{1}\{y_{rit} < 50,000\} \quad \text{sample of a demographic subgroup.} \quad (15)$$

Interpretation. The key requirement is not random sampling in a narrow sense. The key requirement is that the researcher knows enough about the sampling rule to construct the corresponding model analog.

2.4 Micro values, parts, and moments

The paper distinguishes three layers.

- A **micro value** is a scalar function of a sampled consumer:

$$v_{pijt} = v_p(t, j, y_{it}). \quad (16)$$

- A **micro part** is a sample average of micro values:

$$v_p = \frac{1}{N_{d_p}} \sum_{n \in \mathcal{N}_{d_p}} v_{pi_n j_n t_n}. \quad (17)$$

- A **micro moment** is a smooth function of one or more micro parts:

$$f_m(v), \quad v = (v_1, \dots, v_{P_M})'. \quad (18)$$

This structure covers simple averages, ratios of averages, conditional means, covariances, correlations, and regression coefficients.

2.5 Running example: minivans and children

Suppose a survey reports the average number of children among car buyers and the average number of children among minivan buyers. Define

$$v_{1ijt} = \text{kids}_{it}, \quad (19)$$

$$v_{2ijt} = \mathbf{1}\{j \in \mathcal{J}^{\text{mini}}\}, \quad (20)$$

$$v_{3ijt} = \text{kids}_{it} \cdot \mathbf{1}\{j \in \mathcal{J}^{\text{mini}}\}. \quad (21)$$

Then

$$f_1(v) = v_1 \quad (22)$$

matches the average number of children among sampled car buyers, while

$$f_2(v) = \frac{v_3}{v_2} \quad (23)$$

matches the average number of children among minivan buyers.

Economic message. Many common survey statistics are not primitive model objects. They are functions of averages. The paper’s notation makes those functions explicit.

2.6 Empirical example data

The empirical exercise uses Seattle soft-drink demand around the 2018 Seattle sweetened beverage tax. The estimation uses pre-2017 data and asks whether the model predicts observed post-tax substitution patterns.

Key objects:

- NielsenIQ Retailer Scanner data: prices, product characteristics, and sales.
- NielsenIQ Consumer Panel data: household demographics and purchase behavior.
- Seattle data from 2007–2016.
- 2,672 soft-drink UPCs sold at five large retailers.
- 78,161 product-retailer-quarter aggregate observations.
- 40 quarterly markets.
- 10,455 grocery trips with inside soft-drink purchases in the micro data.
- A small Prolific Academic survey that yields 100 second-choice responses from eligible non-diet soft-drink consumers.

Interpretation. The example is not only a policy application. It is a worked example of how to combine scanner data, household panel data, summary moments, and second-choice survey data in one demand estimation problem.

3 Models and empirical specifications

3.1 Mixed-logit demand

Consumer i 's indirect utility from product j in market t is

$$u_{ijt} = \delta_{jt} + \mu_{ijt} + \varepsilon_{ijt}, \quad u_{i0t} = \varepsilon_{i0t}. \quad (24)$$

Mean utility is

$$\delta_{jt} = x'_{jt}\beta + \xi_{jt}. \quad (25)$$

A standard random-coefficients specification is

$$\mu_{ijt} = x'_{jt}(\Pi y_{it} + \Sigma \nu_{it}). \quad (26)$$

Thus consumer i has effective random coefficients

$$\beta_{it} = \beta + \Pi y_{it} + \Sigma \nu_{it}. \quad (27)$$

If idiosyncratic preferences are type I extreme value, the conditional choice probability is

$$s_{ijt} = \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k \in \mathcal{J}_t} \exp(\delta_{kt} + \mu_{ikt})}. \quad (28)$$

Aggregate market share is

$$s_{jt} = \sum_{i \in \mathcal{I}_t} w_{it} s_{ijt}. \quad (29)$$

Interpretation. The matrix Π governs observed demographic heterogeneity. The matrix Σ governs unobserved preference heterogeneity. Micro data are mainly useful because they help identify these heterogeneity parameters.

3.2 Aggregate BLP estimator

Let

$$\theta = (\beta, \Pi, \Sigma). \quad (30)$$

For each candidate (Π, Σ) , BLP inversion finds mean utilities $\hat{\delta}_{jt}(\Pi, \Sigma)$ that match observed market shares:

$$\mathcal{S}_{jt} = s_{jt}(\Pi, \Sigma, \hat{\delta}_t), \quad \forall j \in \mathcal{J}_t. \quad (31)$$

The implied unobserved quality is

$$\hat{\xi}_{jt}(\theta) = \hat{\delta}_{jt}(\Pi, \Sigma) - x'_{jt}\beta. \quad (32)$$

The aggregate sample moments are

$$\hat{g}_A(\theta) = \frac{1}{N_A} \sum_{t \in \mathcal{T}} \sum_{j \in \mathcal{J}_t} \hat{\xi}_{jt}(\theta) z_{jt}. \quad (33)$$

The aggregate BLP estimator solves

$$\hat{\theta}_A = \arg \min_{\theta} \hat{g}_A(\theta)' \hat{W}_A \hat{g}_A(\theta). \quad (34)$$

Interpretation. Aggregate BLP uses instruments to separate price and product quality. But in finite samples, the heterogeneous-taste parameters can be weakly pinned down unless there is strong variation in demographics, prices, and choice sets.

3.3 Why aggregate variation can be weak

The paper uses the linear IV intuition of Salanie and Wolak to explain weak identification of heterogeneous tastes. In the simplest scalar case with one characteristic x_{jt} , one demographic y_{it} , and parameters (β, π, σ) , a second-order approximation gives

$$\log \frac{s_{jt}}{s_{0t}} \approx \beta x_{jt} + \sigma^2 a_{jt} + \pi m_t^y x_{jt} + \pi^2 v_t^y a_{jt} + \xi_{jt}, \quad (35)$$

where

$$a_{jt} = \left(\frac{x_{jt}}{2} - \sum_{k \in \mathcal{J}_t} s_{kt} x_{kt} \right) x_{jt}, \quad (36)$$

and m_t^y and v_t^y are the market-level mean and variance of the demographic.

Identification intuition.

- If m_t^y does not vary across markets, π is difficult to distinguish from β .
- If v_t^y does not vary across markets, π^2 is difficult to distinguish from σ^2 .
- If product assortments barely vary, σ is difficult to identify from aggregate substitution patterns.

Practical advice. Before estimating a highly flexible demand model with only aggregate data, run an approximate IV regression like this one. It helps diagnose whether aggregate data contain enough variation to identify rich heterogeneity.

3.4 Model analogs of micro moments

The model analog of micro part p is a sampling-rule-weighted expectation:

$$v_p(\theta) = \frac{\sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}_t} \sum_{j \in \mathcal{J}_t \cup \{0\}} w_{it} s_{ijt}(\theta) w_{d_p ijt} v_{pijt}}{\sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}_t} \sum_{j \in \mathcal{J}_t \cup \{0\}} w_{it} s_{ijt}(\theta) w_{d_p ijt}}. \quad (37)$$

The model analog of a micro summary statistic is then

$$f_m(v(\theta)). \quad (38)$$

The micro sample moment is

$$\hat{g}_{M,m}(\theta) = f_m(v) - f_m(v(\theta)). \quad (39)$$

Interpretation. Micro moments choose parameters so that the model-predicted relationship

between consumers and choices matches the relationship observed in a survey or consumer panel.

3.5 Combined micro BLP estimator

Stack aggregate and micro moments:

$$\hat{g}(\theta) = \begin{bmatrix} \hat{g}_A(\theta) \\ \hat{g}_M(\theta) \end{bmatrix}, \quad \hat{g}_M(\theta) = \begin{bmatrix} f_1(v) - f_1(v(\theta)) \\ \vdots \\ f_{M_M}(v) - f_{M_M}(v(\theta)) \end{bmatrix}. \quad (40)$$

The estimator solves

$$\hat{\theta} = \arg \min_{\theta} \hat{g}(\theta)' \hat{W} \hat{g}(\theta), \quad \hat{W} = \text{diag}(\hat{W}_A, \hat{W}_M). \quad (41)$$

The weighting matrix is block diagonal when aggregate and micro moments are independent conditional on the aggregate data.

3.6 Nested fixed point with micro moments

For each guess of (Π, Σ) :

1. Solve the BLP share inversion for $\hat{\delta}_{jt}(\Pi, \Sigma)$ in every market.
2. Compute all micro model analogs $f_m(v(\theta))$ using the same individual choice probabilities $s_{ijt}(\theta)$.
3. Concentrate out β by IV/GMM regression of mean utility on characteristics.
4. Form aggregate residuals $\hat{\xi}_{jt}(\theta)$ and aggregate moments $\hat{g}_A(\theta)$.
5. Stack aggregate and micro moments and evaluate the GMM objective.

Interpretation. The micro moments do not replace BLP. They are added to the same nested fixed-point estimation problem.

3.7 Standard micro moments

The paper organizes common moments in the literature into a few types:

$$\mathbb{P}(j \in \mathcal{J}_m \mid i \in \mathcal{I}_m) \quad \text{choice probability for a demographic group,} \quad (42)$$

$$\mathbb{E}[y_{rit} \mid j \in \mathcal{J}_m] \quad \text{average demographic among buyers of a product group,} \quad (43)$$

$$\mathbb{E}[x_{cjt} \mid i \in \mathcal{I}_m, j \neq 0] \quad \text{average product characteristic by demographic group,} \quad (44)$$

$$C(x_{cjt}, y_{rit} \mid j \neq 0) \quad \text{covariance between product characteristics and demographics,} \quad (45)$$

$$C(x_{cjt}, x_{ek(-j)t} \mid j, k \neq 0) \quad \text{covariance between first- and second-choice characteristics.} \quad (46)$$

Economic message. Demographic-choice moments primarily target Π . Second-choice moments primarily target Σ , because they reveal where consumers substitute when their favorite option is

unavailable.

3.8 Second-choice moments

With second-choice data, consumer n has first choice j_n and second choice k_n . The joint choice probability is

$$\mathbb{P}_A(j_n = j, k_n = k \mid t_n = t, i_n = i) = s_{ijkt}. \quad (47)$$

For the logit case,

$$s_{ijkt} = s_{ijt}s_{ik(-j)t}, \quad s_{ik(-j)t} = \frac{s_{ikt}}{1 - s_{ijt}}. \quad (48)$$

The paper also uses the more general expression

$$s_{ijkt} = s_{ik(-j)t} - s_{ikt}, \quad (49)$$

which extends naturally to nested logit-style models.

For a second-choice micro part,

$$v_p(\theta) = \frac{\sum_t \sum_i \sum_j \sum_{k \neq j} w_{it} s_{ijkt}(\theta) w_{d_p i j k t} v_{p i j k t}}{\sum_t \sum_i \sum_j \sum_{k \neq j} w_{it} s_{ijkt}(\theta) w_{d_p i j k t}}. \quad (50)$$

Interpretation. A second choice is like observing a counterfactual choice set for that consumer. This is powerful because the key object in demand estimation is substitution, not only first-choice market share.

3.9 Optimal micro moments

If complete micro data are available, the paper recommends using score-based optimal micro moments in the second GMM step. After a first-stage estimate $\hat{\theta}$, define a micro value for parameter θ_m as

$$v_{mijt}(\hat{\theta}) = \frac{\partial \log \mathbb{P}_A^{\hat{\theta}}(t_n = t, j_n = j, y_{int_n} = y_{it} \mid n \in \mathcal{N}_{d_m})}{\partial \theta_m}. \quad (51)$$

The sample average of this score becomes the target micro moment.

Two-step logic.

1. Estimate the model with standard, interpretable micro moments.
2. Use the first-stage estimate to compute score-based micro values.
3. Re-estimate with optimal instruments, optimal weighting matrix, and score-based micro moments.

Interpretation. Standard moments are intuitive and often data-light. Optimal moments are efficient when complete micro data are available and compatible with the aggregate model.

4 Identification, inference, and credibility

4.1 What identifies each part of the model?

The paper's identification logic is not a single natural experiment. It is an econometric identification and efficiency argument.

- **Mean utilities** are disciplined by aggregate market shares through BLP inversion.
- **Linear mean taste parameters** are disciplined by instruments for endogenous characteristics such as price.
- **Observed heterogeneity** in Π is identified by how demographics predict choices, especially within-market micro relationships such as $C(x_{jt}, y_{it} \mid j \neq 0)$.
- **Unobserved heterogeneity** in Σ is identified by substitution patterns. Aggregate choice-set variation helps, and second-choice data help directly.

Key point. Micro data help most when they contain variation close to the score for the weakly identified parameter.

4.2 Micro data are not automatically valid

The paper emphasizes compatibility. A micro statistic must correspond to the same target population, product definitions, market definitions, and choice environment as the aggregate model.

Common problems include:

- survey respondents face different prices or product assortments;
- the survey is from a different year or geography;
- demographic variables are measured differently;
- survey participation is non-random and weights are ignored;
- the survey underreports the total amount purchased, as in alcohol consumption surveys.

Economic message. A moment that is incompatible with aggregate data does not add information about the same parameter. It adds a conflicting restriction.

4.3 Testing compatibility

If the model can be estimated without the potentially incompatible micro moments, the paper recommends a Wald-style overidentification test. Estimate $\hat{\theta}$ without those micro moments, then compute

$$\hat{\Delta}_M = f(v) - f(v(\hat{\theta})). \quad (52)$$

The test statistic is

$$\text{Wald} = N_A \hat{\Delta}'_M \hat{S}_M^{-1} \hat{\Delta}_M \rightsquigarrow \chi^2(M_M). \quad (53)$$

Interpretation. This checks whether micro statistics look like draws from the population implied by the aggregate model and the moments believed to be valid.

4.4 Inference

The paper derives asymptotic covariance formulas under several thought experiments:

- many markets, including markets covered by micro surveys;
- many aggregate markets with only a few surveyed markets, but large surveys;
- few markets, but many products and large surveys.

A practical implication is important: researchers do not need the survey provider to report sample covariances among summary statistics. Given the sample sizes and the model-implied micro values, PyBLP can compute analytic asymptotic covariance estimators and use the delta method for functions such as ratios and covariances.

4.5 Weighting

The paper separates three types of weights:

- w_{it} : the share of type i consumers in market t in the demand model;
- w_{dijt} : the probability that a consumer of type i choosing j in market t is included in survey d ;
- survey or projection weights: inverse-sampling weights used to make a micro statistic representative of a target population.

Interpretation. Confusing these weights is a common source of mistakes. The model must mimic the statistic actually computed in the micro data.

4.6 Moment selection and diagnostics

The paper recommends using score correlations to choose good moments. Given a preliminary $\hat{\theta}$, simulate micro observations, compute candidate micro values, compute the score contributions, and ask how well the candidate values span the score.

Interpretation. This turns the vague question “which summary statistics are useful?” into a concrete diagnostic: useful moments are highly correlated with the parameter score.

4.7 Numerical integration

The paper extends the integration advice from aggregate BLP.

- Quadrature can be excellent for smooth integrals.
- Monte Carlo or quasi-Monte Carlo is safer when micro moments involve discontinuous indicators or sub-intervals, such as low-income groups.
- Researchers should increase the number of draws or nodes until estimates stabilize.

Key warning. A quadrature rule designed to integrate over a full continuous distribution may perform poorly for moments that condition on an interval such as $y_{it} < \bar{y}_t$.

4.8 Limitations

The paper is a guide to a class of estimators, not a claim that all micro BLP applications are automatically credible.

Important limitations:

- Validity depends on correct model specification and compatible data.
- Second-choice survey responses may contain survey-design biases.
- Optimal moments can be harmful when the micro dataset is incomplete or measured differently.
- Rich models may still be computationally demanding.
- In small samples, adding many moments can trade lower variance for higher finite-sample bias.

5 Tables and figures

5.1 Table 1. Empirical literature

Read: The table lists 29 empirical papers that use micro BLP-style estimation across industries such as automobiles, fast food, personal computers, cigarettes, coffee, health insurance, nursing homes, schools, cereal, rental housing, and retail.

Economic message: Micro BLP is already widely used, but implementations differ across papers. This motivates a common framework and software implementation.

Table 1
Empirical literature.

Paper	Demand estimation	Country	Years
	Industry		
Petrin (2002)	Automobiles	United States	1981–1993
Berry et al. (2004a)	Automobiles	United States	1993
Thomsen (2005)	Fast Food	United States	1999
Goeree (2008)	Personal Computers	United States	1996–1998
Ciliberto and Kuminoff (2010)	Cigarettes	United States	1993–2002
Nakanura and Zeron (2010)	Coffee	United States	2000–2004
Beresteanu and Li (2011)	Automobiles	United States	1999–2006
Li (2012)	Automobiles	United States	1999–2006
Copeland (2014)	Automobiles	United States	1999–2008
Starc (2014)	Health Insurance	United States	2004–2008
Ching et al. (2015)	Nursing Homes	United States	1999
Li et al. (2015)	Automobiles	China	2004–2009
Nurski and Verboven (2016)	Automobiles	Belgium	2010–2011
Murry (2017)	Automobiles	United States	2007–2011
Wollmann (2018)	Commercial Vehicles	United States	1986–2012
Li (2018)	Automobiles	China	2008–2012
Li et al. (2018)	Digital Cameras	United States	2007–2010
Backus et al. (2021)	Cereal	United States	2007–2016
Barwick et al. (2021)	Automobiles	China	2009–2011
Nelson (2021)	Primary Schools	Chile	2005–2016
Armitage and Pinter (2022)	Automobiles	United States	2009–2017
Durmeyer (2022)	Automobiles	France	2003–2008
Weber (2022)	Trucks	United States	2010–2018
Bodéré (2023)	Preschools	United States	2010–2018
Grieco et al. (2023b)	Automobiles	United States	1980–2018
Calder-Wang and Kim (2024)	Rental Housing	United States	2011–2018
Gonlon and Rao (2024)	Distilled Spirits	United States	2007–2013
Döpper et al. (2024)	Retail	United States	2006–2019
Montag (2024)	Laundry Machines	United States	2005–2015

This table collects a non-exhaustive list of empirical papers that use the micro BLP estimator, along with the industry, country, and years for which each paper estimates demand. Some papers estimate demand for the listed broad industry and subsequently focus on a sub-industry. We only list published and recent working papers that do not diverge too much from the standard demand-side BLP model. In Table 3 we reorganize these papers by which micro moments they use.

Table 2
Notation.

Notation for aggregate data and estimation (Section 2)		Notation for micro data and estimation (Section 4)	
$t \in \mathcal{T}$	Markets	$d \in \mathcal{D}$	Micro datasets
$\mathcal{M}_t \in \mathbb{R}_+$	Market size	$w_{d it} \in [0, 1]$	Sampling probability
		$w_{d it} \in [0, 1]$	Joint sampling probability
$j \in \mathcal{J}_t$	Products	$n \in \mathcal{N}_d^j$	Micro observations
$j = 0$	Outside alternative	$t_s \in \mathcal{T}$	Micro observation market
$c = 1, \dots, C$	Observed product characteristics	$i_s \in \mathcal{I}_{t_s}$	Micro observation type
$m = 1, \dots, M_A$	Instruments	$j_s \in \mathcal{J}_{t_s} \cup \{0\}$	Micro observation choice
$x_{ijt} \in \mathbb{R}$	Observed product characteristic	$k_s \in \mathcal{J}_{t_s} \cup \{0\} \setminus \{j_s\}$	Micro observation second choice
$x_{ijt} \in \mathbb{R}^{n \times 1}$	All observed product characteristics		
$z_{ijt} \in \mathbb{R}$	Instrument	$p = 1, \dots, P_d$	Micro parts
$z_{ijt} \in \mathbb{R}^{M_A \times 1}$	All instruments	$d_p \in \mathcal{D}$	Micro part dataset
$\zeta_{it} \in \mathbb{R}$	Mean-zero unobserved product quality	$v_{p it} \in \mathbb{R}$	Micro part value
		$v_{p it} \in \mathbb{R}$	Second choice micro part value
$i \in \mathcal{I}_t$	Consumer types	$m = 1, \dots, M_d$	Micro moments
$r = 1, \dots, R$	Consumer demographics	$f_m : \mathbb{R}^{M_A \times 1} \rightarrow \mathbb{R}$	Micro moment function
$w_{it} \in [0, 1]$	Consumer type share		
$y_{it} \in \mathbb{R}$	Consumer demographic	$\bar{v}_j \in \mathbb{R}$	Micro part sample value
$y_{it} \in \mathbb{R}^{R \times 1}$	All consumer demographics	$\bar{v} \in \mathbb{R}^{P_d \times 1}$	All micro part sample values
$v_{it} \in \mathbb{R}$	Unobserved preference	$f_p(\bar{v}) \in \mathbb{R}$	Micro moment sample value
$v_{it} \in \mathbb{R}^{n \times 1}$	All unobserved preferences		
$u_{it} \in \mathbb{R}$	Indirect utility	$v_p(\theta) \in \mathbb{R}$	Micro part expected value
$\delta_{it} \in \mathbb{R}$	Mean utility	$v(\theta) \in \mathbb{R}^{P_d \times 1}$	All micro part expected values
$\mu_{it} \in \mathbb{R}$	Heterogeneous utility	$f_p(v(\theta)) \in \mathbb{R}$	Micro moment expected value
$\epsilon_{ijt} \in \mathbb{R}$	Idiosyncratic preference		
$s_{ijt} \in (0, 1)$	Choice probability	$s_{k it} \in (0, 1)$	Joint choice probability
$s_{jt} \in (0, 1)$	Market share	$s_{k it} \in (0, 1)$	Probability of choosing k without j
$S_{jt} \in (0, 1)$	Observed market share	$s_{k it} \in (0, 1)$	The same, but without a group $h(j)$
$\beta \in \mathbb{R}^{C \times 1}$	Linear parameters	$M = M_A + M_d$	Number of combined moments
$\Pi \in \mathbb{R}^{C \times R}$	Consumer demographic parameters	$\hat{g}(\theta) \in \mathbb{R}^{M \times 1}$	Combined sample moments
$\Sigma \in \mathbb{R}^{C \times C}$	Unobserved preference parameters	$\hat{W} \in \mathbb{R}^{M \times M}$	Combined weighting matrix
$\theta = (\beta, \Pi, \Sigma)$	All parameters		
$N_A = \sum_{t \in \mathcal{T}} \mathcal{J}_t $	Number of aggregate observations	$N_d = \mathcal{N}_d^j $	Number of micro observations
$\hat{g}_A(\theta) \in \mathbb{R}^{M_A \times 1}$	Aggregate sample moments	$\hat{g}_d(\theta) \in \mathbb{R}^{M_d \times 1}$	Micro sample moments
$\hat{W}_A \in \mathbb{R}^{M_A \times M_A}$	Aggregate weighting matrix	$\hat{W}_d \in \mathbb{R}^{M_d \times M_d}$	Micro weighting matrix

This table summarizes the notation we introduce in Sections 2 and 4. Subscripts on parameters such as θ_0 refer to true values. Subscripts on operators such as P_A indicate conditioning on all aggregate data.

5.2 Table 2. Notation

Read: The table collects aggregate-data notation on the left and micro-data notation on the right. It defines markets, products, characteristics, consumer types, mean utilities, micro datasets, micro parts, micro moments, and combined GMM moments.

Economic message: Much of the contribution is organizational. By making all objects explicit, the paper allows researchers to compare different micro moments within the same estimator.

5.3 Table 3. Micro moment examples

Read: The table classifies moments used in the literature into five broad types: conditional choice probabilities, demographics conditional on product groups, product characteristics conditional on demographics, characteristic-demographic covariances, and second-choice covariances.

Economic message: Most empirical micro BLP moments are special cases of smooth functions of sample averages.

Table 3
Micro moment examples.

Shorthand	Papers
" $\Pi_j \in \mathcal{J}_x, \forall j \in \mathcal{J}_x$ "	Petrin (2002), Thomadsen (2005), Goeree (2006), Nakamura and Zenmu (2010), Berezanski and Li (2011), Li (2012), Stanc (2014), Ching et al. (2015), Li et al. (2015), Barwick et al. (2021), Li (2018), Li et al. (2018), Boddé (2023)
" $\mathbb{E}(I_{j,i} j \in \mathcal{J}_x)$ "	Petrin (2002), Ciliberto and Kuminoff (2010), Li (2012), Copeland (2014), Nurski and Verboven (2016), Murray (2017), Wolfsen (2018), Backus et al. (2021), Armitage and Pfarrer (2022), Süßner et al. (2024), Durnmeyer (2022), Weber (2022), Conlon and Rao (2024)
" $\mathbb{E}(x_{j,i} i \in \mathcal{J}_x, j \neq 0)$ "	Stanc (2014), Grieco et al. (2023b), Neilson (2021), Weber (2022), Boddé (2023), Conlon and Rao (2024)
" $C(x_{j,i}, I_{j,i} j \neq 0)$ "	Berry et al. (2004a), Nurski and Verboven (2016), Backus et al. (2021), Durnmeyer (2022), Montag (2024), Calder-Wang and Kim (2024)
" $C(x_{j,i}, x_{k,i} j, k \neq 0)$ "	Berry et al. (2004a), Grieco et al. (2023b), Montag (2024)

This table lists examples of micro moments that we discuss in Section 5. Each row lists our notation-abusing shorthand and empirical papers from Table 1 that have used essentially the same micro moment.

Table 4
Demographic variation.

Variation	Distributions	Markets	MAE (%)					Bias (%)				
			$\hat{\pi}_1$	$\hat{\pi}_x$	$\hat{\beta}_1$	$\hat{\beta}_x$	$\hat{\delta}$	$\hat{\pi}_1$	$\hat{\pi}_x$	$\hat{\beta}_1$	$\hat{\beta}_x$	$\hat{\delta}$
National	1	40	436.6	133.3	8.3	5.1	1.2	-110.3	-45.6	1.8	2.1	0.1
States	50	40	197.8	60.6	3.9	2.4	1.2	-31.3	-12.6	0.6	0.4	0.1
PUMAs	982	40	97.5	30.0	2.7	1.4	1.2	-5.8	-4.7	0.2	0.2	0.1
National	1	80	327.7	102.8	6.1	4.0	0.9	-98.5	-48.4	2.0	2.1	0.0
States	50	80	139.5	42.7	2.7	1.6	0.9	-7.4	-6.1	0.2	0.2	-0.0
PUMAs	982	80	65.9	21.3	1.9	1.0	0.8	-4.9	-2.4	0.2	0.2	-0.0

This table reports median absolute error (MAE) and median bias of parameter estimates over 1,000 simulated datasets for different amounts of cross-market demographic variation. We randomly assign each market either to the same national distribution of income, to one of 50 US states, or to one of the 982 Public Use Microdata Areas (PUMAs) used by the American Community Survey (ACS). In the last three rows, we simulate 40 more markets, keeping the same demographic distributions as in the first 40, but with different choice sets.

5.4 Table 4. Demographic variation

Read: In the Monte Carlo, aggregate-only estimation performs poorly when demographic variation is weak. With 40 markets, the MAE for $\hat{\pi}_1$ is 436.6 under a single national income distribution, 197.8 with state-level income distributions, and 97.5 with PUMA-level distributions. Doubling to 80 markets improves performance but does not eliminate the problem.

Economic message: Aggregate data can identify demographic heterogeneity only if demographics and choice sets vary meaningfully across markets. Otherwise, the estimator is noisy even with good instruments.

5.5 Table 5. Standard micro moments

Read: With no micro moments, the MAE is 197.8 for $\hat{\pi}_1$ and 60.6 for $\hat{\pi}_x$. Matching only $\mathbb{E}[y_{it} | j \neq 0]$ improves these to 164.8 and 44.9. Matching only $C(x_{2jt}, y_{it} | j \neq 0)$ improves them to 53.8 and 11.7. Matching both gives 34.0 and 10.8.

Economic message: A well-targeted covariance between demographics and product characteristics is much more informative than a coarse demographic mean alone. Combining complementary moments works best.

5.6 Figure 1. Standard micro moment correlations

Read: The figure compares correlations between candidate micro statistics and the score for π_x . The continuous interaction $x_{2jt}y_{it}$ is strongly correlated with the score, about 0.675. Discretizing income lowers the correlation to about 0.6.

Economic message: Score correlation explains why some micro moments are more powerful than others. Discretization can make a statistic easier to collect but less informative.

Micro moments shorthand	MAE (%)		Bias (%)	
	$\hat{\theta}_1$	$\hat{\theta}_x$	$\hat{\theta}_1$	$\hat{\theta}_x$
No Micro Moments	197.8	60.6	-31.3	-12.6
" $E[y_{it} j \neq 0]$ "	164.8	44.9	2.9	1.4
" $C(x_{2it}, y_{it} j \neq 0)$ "	53.8	11.7	17.5	2.3
" $E[y_{it} j \neq 0, C(x_{2it}, y_{it} j \neq 0)]$ "	34.0	10.8	4.1	1.1
" $E[y_{it} j \neq 0, E(x_{2it}, y_{it} j \neq 0)]$ "	37.6	12.0	3.2	0.7
" $E[y_{it} j \neq 0, E(x_{2it} y_{it} < \bar{y}_j, j \neq 0)]$ "	62.7	17.3	0.9	1.0
" $E[y_{it} j \neq 0, E(x_{2it} y_{it} < \bar{y}_j, j \neq 0), C(x_{2it}, y_{it} j \neq 0)]$ "	34.2	10.7	4.3	1.1

This table reports median absolute error (MAE) and median bias of parameter estimates over 1,000 simulated datasets for different combinations of standard micro moments. The cutoff $\bar{y}_j$ is the median income y_{it} in market t .

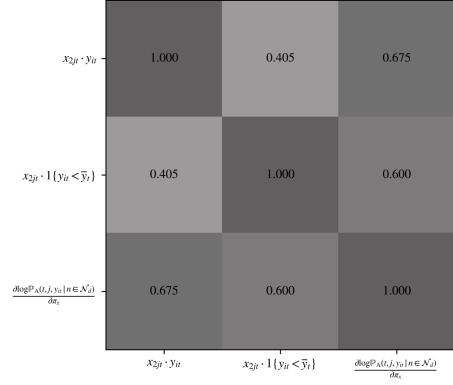


Fig. 1. Standard Micro moment correlations. This figure reports median absolute correlations between different micro statistics over 1,000 simulated micro datasets underlying the micro moments in Table 5. For each micro observation s in market $t = t$ of type $i_s = i$ with choice $j_s = j$, we compute three statistics: $x_{2it} \cdot y_{it}$ captures variation in " $E(x_{2it}, y_{it} | j \neq 0)$ " and " $C(x_{2it}, y_{it} | j \neq 0)$ " moments, $x_{2it} \cdot 1\{y_{it} < \bar{y}_j\}$ captures variation in " $E(x_{2it} | y_{it} < \bar{y}_j, j \neq 0)$ " moments, and $\frac{\partial \log P_A(i, j, y_{it} | n \in N'_j)}{\partial \theta_x}$ is the score for θ_x at the true θ_0 .

5.7 Table 6. Optimal micro moments and compatibility

Read: Using standard covariance moments gives MAE 34.0 for $\hat{\pi}_1$ and 10.8 for $\hat{\pi}_x$. Replacing them in the second step with optimal score moments improves the MAEs to 23.8 and 6.3. However, when the micro dataset is incompatible because true income is replaced by a censored income measure, optimal moments can perform badly: the MAE for $\hat{\pi}_1$ rises to 107.1.

Economic message: Optimal moments are optimal only when the underlying micro data are complete and compatible with the model. With incompatible data, robust simple statistics can dominate score-based moments.

Micro moments (plus $E[y_{it} j \neq 0]$)	Incompatible	Optimal	MAE (%)		Bias (%)	
			$\hat{\theta}_1$	$\hat{\theta}_x$	$\hat{\theta}_1$	$\hat{\theta}_x$
" $C(x_{2it}, y_{it} j \neq 0)$ "			34.0	10.8	4.1	1.1
" $C(x_{2it}, y_{it} j \neq 0)$ "		Yes	23.8	6.3	-0.6	-0.1
" $E(x_{2it} y_{it} < \bar{y}_j, j \neq 0)$ "			62.7	17.3	0.9	1.0
" $E(x_{2it} y_{it} < \bar{y}_j, j \neq 0)$ "		Yes	24.1	6.4	-0.5	-0.4
" $E(x_{2it} y_{it} < \bar{y}_j, j \neq 0)$ "	Yes		64.4	18.0	0.8	0.7
" $E(x_{2it} y_{it} < \bar{y}_j, j \neq 0)$ "	Yes	Yes	107.1	19.1	104.6	-13.7

This table reports median absolute error (MAE) and median bias of parameter estimates over 1,000 simulated datasets for standard and optimal micro moments. The first and third rows are the same as the fourth and sixth rows in Table 5. The second and fourth rows use these same standard micro moments in the first GMM step to construct optimal micro moments for the second step. For the last two rows, we simulate a second, independent micro dataset that is configured the same, except we replace y_{it} with $\bar{y}_j$: the 25th percentile of income if below the median or the 75th percentile if above. We use this second dataset for " $E(x_{2it} | y_{it} < \bar{y}_j, j \neq 0)$ " and in the last row, optimal micro moments as well.

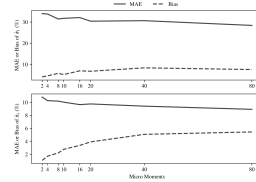


Fig. 2. Pooling markets. This figure reports median absolute error (MAE) and median bias of parameter estimates over 1,000 simulated datasets for an increasing number of micro moments. The top graph shows MAE and median bias for $x_{2it} \cdot y_{it}$ and $x_{2it} \cdot 1\{y_{it} < \bar{y}_j\}$. The bottom graph shows MAE and median bias for $\frac{\partial \log P_A(i, j, y_{it} | n \in N'_j)}{\partial \theta_x}$. The x-axis is the number of micro moments (1 to 100). The y-axis is the MAE and Bias. The legend indicates MAE (solid line) and Bias (dashed line).

5.8 Figure 2. Pooling markets

Read: The figure varies how much micro moments are pooled across markets. More market-specific micro moments reduce variance but can add finite-sample bias.

Economic message: There is a many-moments tradeoff. Researchers should not mechanically create a separate micro moment for every market if that produces too many noisy restrictions.

5.9 Table 7. Numerical integration

Read: For smooth covariance moments, 7-node Gauss–Hermite quadrature performs similarly to 1,000 Monte Carlo draws: MAE 31.6 and 9.2 versus 34.0 and 10.8. But for a moment with a low-income indicator, quadrature performs very poorly: MAE 251.8 and 71.0 versus 62.7 and 17.3 with Monte Carlo.

Economic message: Quadrature is efficient for smooth integration, but it can fail for moments that condition on discrete or interval-defined subgroups.

Table 7

Numerical integration.

Micro moments (plus “ $E[y_{it} j \neq 0]$ ”)	Integration	MAE (%)		Bias (%)	
		$\hat{\pi}_1$	$\hat{\pi}_2$	$\hat{\pi}_1$	$\hat{\pi}_2$
“ $C(x_{jit}, y_{it} j \neq 0)$ ”	Quadrature	31.6	9.2	−1.9	−1.1
“ $C(x_{jit}, y_{it} j \neq 0)$ ”	Monte Carlo	34.0	10.8	4.1	1.1
“ $E[x_{jit} y_{it} < \bar{y}_i, j \neq 0]$ ”	Quadrature	251.8	71.0	24.6	5.4
“ $E[x_{jit} y_{it} < \bar{y}_i, j \neq 0]$ ”	Monte Carlo	62.7	17.3	0.9	1.0

This table reports median absolute error (MAE) and median bias of parameter estimates over 1,000 simulated datasets for different choices of consumer types I_i for numerically integrating over the lognormal population distribution of income y_{it} . “Quadrature” refers to $|I_i| = 7$ Gauss–Hermite nodes and weights that exactly integrate polynomials of degree $2 \times 7 - 1 = 13$ or less. Quadrature nodes are transformed into nodes for income with the mean and standard deviation of log income in each market. “Monte Carlo” refers to $|I_i| = 1,000$ pseudo-Monte Carlo draws from the true distribution of income. The cutoff $\bar{y}_i$ is the median income y_{it} in market i .

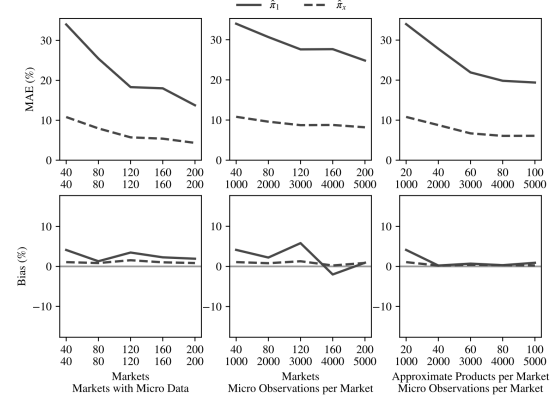


Fig. 3. Problem scaling.

This figure reports median absolute error (MAE) and median bias of parameter estimates over 1,000 simulated datasets as finite sample sizes approach the three asymptotic thought experiments discussed in Appendix E. In all panels we match the same “ $E[y_{it} | j \neq 0]$ ” and “ $C(x_{jit}, y_{it} | j \neq 0)$ ” moments in the fourth row of Table 5. The leftmost panel fixes the number of products and micro observations per market and scales up the number of markets, including those with micro data. The middle panel fixes the number of products per market and the number of markets with micro data and scales up the number of aggregate markets and the number of micro observations in each of the fixed number of markets. The rightmost panel fixes the number of markets and scales up the number of products and micro observations per market.

5.10 Figure 3. Problem scaling

Read: The figure studies finite-sample performance under three asymptotic thought experiments: many markets with micro data, many aggregate markets but few surveyed markets with large surveys, and few markets with many products and large surveys.

Economic message: The estimator behaves sensibly as sample sizes grow. The variance of estimates declines under all three scaling regimes.

5.11 Table 8. Unobserved heterogeneity

Read: When choice sets vary across markets, unobserved heterogeneity σ_x is estimated well even without micro moments. When all markets have the same choice set, the MAE for $\hat{\sigma}_x$ is about 99.5 without micro moments, and standard demographic micro moments do not fix the problem.

Economic message: Micro demographic moments are not enough to identify unobserved heterogeneity. To learn Σ , the researcher needs aggregate choice-set variation, second-choice data, or other data that directly reveal substitution patterns.

5.12 Table 9. Second choices

Read: In the no-choice-set-variation case, without second-choice moments the MAE for $\hat{\sigma}_x$ is 94.4. A second-choice covariance reduces it to 16.4. A second-choice sum moment reduces it to 5.3.

Table 8
Unobserved heterogeneity.

Micro moments shorthand	$J_i = J$	Optimal	MAE (%)			Bias (%)		
			$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\theta}_3$	$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\theta}_3$
No Micro Moments			225.7	76.5	3.4	-43.4	-14.6	-0.3
"E[y _{ij} j ≠ 0], C(x _{2ij} , y _j j ≠ 0)"			39.2	12.1	3.2	3.3	0.1	-0.3
"E[y _{ij} j ≠ 0], C(x _{2ij} , y _j j ≠ 0)"		Yes	29.1	8.3	3.3	-2.8	-0.7	-0.3
No Micro Moments		Yes	153.6	79.3	99.5	2.8	21.9	31.8
"E[y _{ij} j ≠ 0], C(x _{2ij} , y _j j ≠ 0)"		Yes	33.6	23.1	94.0	-0.8	-13.8	-82.3
"E[y _{ij} j ≠ 0], C(x _{2ij} , y _j j ≠ 0)"		Yes	31.8	24.0	99.2	-5.5	-17.9	-86.4

This table reports median absolute error (MAE) and median bias of parameter estimates over 1,000 simulated datasets with unobserved preferences for different amounts of choice set variation and different micro moments. We draw unobserved preferences v_{ij} from the standard normal distribution and add $\sigma_{x_{2ij}}v_{2ij}$ to μ_{ij} with the true $\sigma_{v_{ij}} = 0.5$. In the bottom three rows, we use the same choice set $J_i = J$ in each market, cluster our estimates of the asymptotic covariance matrix for $\hat{\theta}_{ij}$ by product j , and use the number of markets T as the number of aggregate observations N_{ij} .

Table 9

Micro Moments (plus "E[y _{ij} j ≠ 0], C(x _{2ij} , y _j j ≠ 0)"]	Optimal	MAE (%)			Bias (%)		
		$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\theta}_3$	$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\theta}_3$
No Second Choice Moments		33.2	23.2	94.4	-0.9	-14.0	-82.7
"C(x _{2ij} , y _{2ij} j, k ≠ 0)"		33.9	12.0	16.4	3.7	0.2	-2.3
"E(x _{2ij} + x _{2ij} j, k ≠ 0)"		34.0	10.4	5.3	4.4	1.7	-0.5
"P(x _{2ij} < x _{2j} x _{2ij} < x _{2j} , j, k ≠ 0)"		34.7	11.0	12.5	3.5	1.9	-2.9
"P(x _{2ij} < x _{2j} x _{2ij} < x _{2j} , j, k ≠ 0)"	Yes	16.9	4.8	4.3	-0.3	-0.8	-1.0

This table reports median absolute error (MAE) and median bias of parameter estimates over 1,000 simulated datasets with unobserved preferences for different second choice moments based on second choice data. We draw unobserved preferences v_{ij} from the standard normal distribution and add $\sigma_{x_{2ij}}v_{2ij}$ to μ_{ij} with the true $\sigma_{v_{ij}} = 0.5$. To eliminate cross-market choice set variation, we use the same choice set $J_i = J$ in each market, cluster our estimates of the asymptotic covariance matrix for $\hat{\theta}_{ij}$ by product j , and use the number of markets T as the number of aggregate observations N_{ij} . In addition to the main micro dataset, we simulate a second, independent micro dataset that is configured the same, except that it also reports second choices. The shorthand "P(x_{2ij} < x_{2j} | x_{2ij} < x_{2j}, j, k ≠ 0)" refers to two moments that match the share of individuals who divert from a below- or above-median x_{2ij} first choice j to a below-median x_{2ij} second choice k .

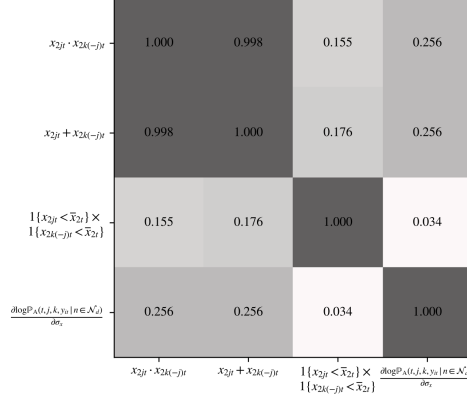


Fig. 4. Second choice micro moment correlations.
This figure reports median absolute correlations between different micro statistics over 1,000 simulated micro datasets underlying the second choice moments in Table 9. For each micro observation n in market i of type $i_k = i$ with choices $j_i = j$ and $k_i = k$, we compute four statistics: $x_{2ij} \cdot x_{2k(-j)j}$ captures variation in "C(x_{2ij}, x_{2k(-j)j} | j, k ≠ 0)" moments, $x_{2ij} + x_{2k(-j)j}$ captures variation in "E(x_{2ij} + x_{2k(-j)j} | j, k ≠ 0)" moments, $1\{x_{2ij} < \bar{x}_{2j}\} \cdot 1\{x_{2k(-j)j} < \bar{x}_{2j}\}$ captures variation in "P(x_{2ij} < x_{2j} | x_{2ij} < x_{2j}, j, k ≠ 0)" moments, and $d \log P_x(i,j,k,y_i | n \in N_i) / d \sigma_i$ is the score for σ_i at the true θ_i .

Optimal second-choice moments reduce it further to 4.3.

Economic message: Second choices are extremely valuable because they directly observe substitution. They can identify unobserved preference heterogeneity even when aggregate choice sets barely vary.

5.13 Figure 4. Second-choice micro moment correlations

Read: The figure compares second-choice statistics to the score for σ_x . Statistics based on first- and second-choice characteristics are highly informative about unobserved taste heterogeneity.

Economic message: A second-choice survey should be designed around the parameters that matter for the policy counterfactual. If the key uncertainty is substitution to outside options or diet products, ask directly about those substitutions.

5.14 Table 10. Seattle sweetened beverage tax

Read: The aggregate specification estimates a weighted average taxed elasticity around -1.35 and predicts about a 30 percent decline in taxed volume with little substitution to untaxed drinks. Adding standard micro moments gives very similar aggregate predictions but allows heterogeneous predictions by income and children. Adding second-choice diversion moments changes the substitution pattern: taxed volume falls by about 15.9 percent and untaxed volume rises by about 9.2 percent. The optimal-moment specification is similar to the diversion specification.

Economic message: Standard micro moments discipline demographic heterogeneity, but second-choice moments discipline substitution to the outside good and to diet drinks. This materially

Table 10
Predicting substitution from seattle's sweetened beverage tax.

		Aggregate	Micro moments		
			Standard	Diversion	Optimal
Price/Once Coefficients	Constant	-52.645 (4.660)	-52.343 (4.694)	-38.538 (4.034)	-37.902 (4.217)
	High Income Household		3.549 (0.940)	3.178 (1.046)	4.062 (0.992)
	Child in Household		-6.915 (1.274)	-8.119 (1.406)	-11.105 (1.458)
	Unobserved Preference	19.631 (1.802)	19.229 (1.805)	15.256 (2.569)	14.941 (2.749)
Inside Goods Coefficients	High Income Household		-0.053 (0.040)	0.346 (0.130)	-0.278 (0.120)
	Child in Household		0.498 (0.050)	1.210 (0.229)	1.884 (0.352)
	Unobserved Preference		4.964 (0.387)	5.378 (0.410)	5.178 (0.410)
	High Income Household		0.708 (0.043)	0.999 (0.142)	0.684 (0.124)
Diet Formula Coefficients	Child in Household		-0.852 (0.056)	-1.463 (0.271)	-1.037 (0.216)
	Unobserved Preference		2.606 (0.868)	2.071 (0.950)	2.071 (0.950)
Small Sized Coefficients	High Income Household		-0.690 (0.060)	-0.710 (0.061)	-0.662 (0.058)
	Child in Household		0.689 (0.069)	0.716 (0.071)	0.641 (0.066)
Standard Micro Statistics	" $\Psi(\text{High}_i, j \neq 0) = 0.337$ "		0.597 (0.202)	0.597 (0.202)	0.561 (0.228)
	" $\Psi(\text{Child}_i, j \neq 0) = 0.203$ "		-0.0004 (0.0001)	-0.0004 (0.0001)	-0.0002 (0.0004)
	" $\Psi(\text{Price}_i, \text{High}_i, j \neq 0) = -0.0004$ "		-0.0001 (0.0355)	-0.0001 (0.0355)	-0.0004 (0.0220)
	" $\Psi(\text{Price}_i, \text{Child}_i, j \neq 0) = -0.0001$ "		-0.0264 (0.0207)	-0.0264 (0.0207)	-0.0172 (0.0192)
	" $\Psi(\text{Diet}_i, \text{High}_i, j \neq 0) = 0.0355$ "		0.0116 (0.0116)	0.0116 (0.0116)	0.0089 (0.0089)
	" $\Psi(\text{Diet}_i, \text{Child}_i, j \neq 0) = -0.0264$ "		0.92 (0.03)	0.93 (0.03)	0.14 (0.17)
	" $\Psi(\text{Diet}_{i,j,i} = 0 \text{Surveyed Non-diet}_{i,j}) = 0.17$ "		78.161 (2672)	78.161 (2672)	78.161 (2672)
	Product-Retailer Quarters		5 (40)	5 (40)	5 (40)
	Product-Retailers		425 (5815)	425 (5815)	425 (5815)
	Product-Retailer-Quarters		200 (10,455)	200 (10,455)	200 (10,455)
	Grocery Trips		1,130 (10)	1,130 (10)	1,130 (10)
	Household Years		100 (100)	100 (100)	100 (100)
Second Choice Responses	Weighted Average Taxed Elasticity	-1.354 (0.064)	-1.349 (0.065)	-1.327 (0.090)	-1.320 (0.095)
	Taxed Volume Change (%)	-30.095 (1.439)	-29.973 (1.452)	-15.870 (1.679)	-15.652 (1.487)
	Low Income Households		-31.134 (1.435)	-16.472 (1.941)	-16.289 (1.626)
	High - Low Income		1.967 (0.778)	1.026 (0.895)	1.304 (0.615)
	Households without Children		-28.790 (1.471)	-15.282 (1.760)	-14.508 (1.626)
	With - without Children		-4.891 (0.915)	-2.475 (1.492)	-4.793 (2.218)
	Untaxed Volume Change (%)	0.872 (0.036)	0.835 (0.039)	9.238 (3.430)	9.383 (3.490)
	Low Income Households		0.948 (0.040)	12.762 (4.000)	11.643 (3.919)

(continued on next page)

Table 10 (continued).

	Aggregate	Micro moments		
		Standard	Diversion	Optimal
↔ High - Low Income		-0.163 (0.039)	-5.114 (1.370)	-3.522 (0.850)
↔ Households without Children		0.736 (0.035)	8.096 (3.256)	8.453 (3.246)
↔ With - without Children		0.620 (0.041)	7.234 (1.540)	4.740 (1.553)

This table reports results for the empirical example described in Section 8. From left to right, we report estimates using aggregate moments, adding standard micro moments, adding second choice moments, and replacing standard micro moments with the optimal micro moments described in Section 6. Standard errors are clustered by brand for the aggregate moments and are in parentheses; we compute those for tax counterfactual with a parametric bootstrap.

changes the tax counterfactual.

5.15 Table 10 continued. Data and implementation details

Read: The empirical model includes product-retailer and retailer-quarter fixed effects, clusters standard errors by brand, uses 10,455 grocery trips, and adds 100 second-choice survey responses.

Economic message: The example shows how a small targeted survey can be combined with large scanner and panel datasets to improve a structural counterfactual.

6 Short summary

This paper extends recommended practices for BLP estimation to the case with micro data. The estimator stacks aggregate BLP moments with micro moments that compare observed micro statistics to model-predicted micro statistics. The framework is broad enough to include simple averages, conditional means, covariances, weighted survey statistics, second-choice moments, and optimal score moments.

The main econometric lesson is that micro data are most valuable when aggregate variation is weak and when the micro statistic is closely related to the score for the parameter of interest. Demographic-choice moments help estimate observed heterogeneity Π . Second-choice moments help estimate unobserved heterogeneity Σ . Optimal micro moments can improve efficiency, but only when the micro data are complete and compatible with the aggregate model.

The main practical lesson is to be disciplined. Define the survey population, sampling weights, micro values, and moment functions. Check compatibility. Use score correlations to choose informative

moments. Use quadrature only when the moment is smooth. Be cautious about many micro moments. When possible, collect targeted second-choice data for the substitution patterns that matter in the policy counterfactual.

7 Conclusion

7.1 Core conclusion

Incorporating micro data into BLP-style demand estimation can substantially improve finite-sample performance and counterfactual credibility, especially when aggregate data alone do not contain enough variation to estimate rich substitution patterns.

7.2 Economic intuition

Aggregate shares tell us which products are popular. Micro data tell us who buys what and where consumers would go if their favorite option disappeared. The first source identifies average demand; the second source disciplines heterogeneity and substitution.

7.3 Methodological takeaway

The paper turns micro BLP from a collection of ad hoc empirical implementations into a standardized minimum-distance framework. The same notation covers survey summaries, panels, second-choice data, and optimal score moments.

7.4 Applied takeaway

For empirical researchers, the recommended workflow is:

1. estimate or diagnose the aggregate BLP model;
2. identify which heterogeneity parameters are weakly estimated;
3. choose micro moments that target those parameters;
4. verify compatibility between aggregate and micro data;
5. use optimal moments when complete compatible micro data are available;
6. design second-choice surveys when substitution patterns are central to the policy question.

7.5 Final takeaway

The paper's message is not simply "add micro data." The message is: add the right micro data, define the exact statistic being matched, weight it consistently, test whether it is compatible, and use it to discipline the substitution patterns that the counterfactual requires.